A few remarks on meta-analyses

Witold Więcek

15 November 2021

History

- Cochran (1953): averaging means via inverse-variance weighting
  - Building up from Cochran and Fisher ideas dating back to 1930s
- 1970-1990: RCTs as basis of clinical decision-making and “evidence-based” medicine (Archie Cochrane)
- Glass (1976): Primary, secondary, and meta-analysis of research
  - First time term meta-analysis used
  - Many estimation methods established throughout 1980s
- Rubin (1981): Estimation in parallel experiments
  - Bayesian hierarchical model
  - Much rapid advance only from 1990s due to computational advances allowing for MCMC
- Exponential growth in number of meta-analyses since 1990s
  - Cochrane Collaboration (1993) and Campbell Collaboration (2000): systematic reviews in health and beyond
  - Guidelines for meta-analyses also developed in 1990s

Critiques

- Researcher degrees of freedom in (1) choice of studies, (2) building models
- Meta-analyses make it easy to ignore or underweight non-RCT evidence
- “GIGO” issues
  - Meta-analysis of low quality (small) studies can lead to overconfidence
  - Risk of giving extra credibility to poor research
  - Statistical power issues also apply to meta-analyses
- Institutional critiques
  - John Ioannidis, 2016: 3% of meta-analyses are “decent and clinically useful” (speculative)
  - Richard Horton (editor in chief, the Lancet), 2019: “Imagine if the entire edifice of knowledge in medicine was built upon a falsehood.”
  - Gøtzsche (BMJ EBM), 2019: “The Cochrane HPV vaccine review was incomplete and ignored important evidence of bias” (only 26out of 46 trials included, CoI)

Choice of studies (quick summary)

- Preregistration (e.g. Cochrane Library)
- PRISMA (checklist & flow diagram) prisma-statement.org
- Assessing quality of studies, risk of bias
  - Bias: selection, attrition, performance, detection, within-study reporting
- Assessing (across-study) publication bias

Choice of outcome

- Continuous: differences between means, but how to standardise?
  - Choice not only of outcome but also of an estimator
- Binary: risk difference, risk ratio, odds ratio
- Other types of data and statistics can also be meta-analysed

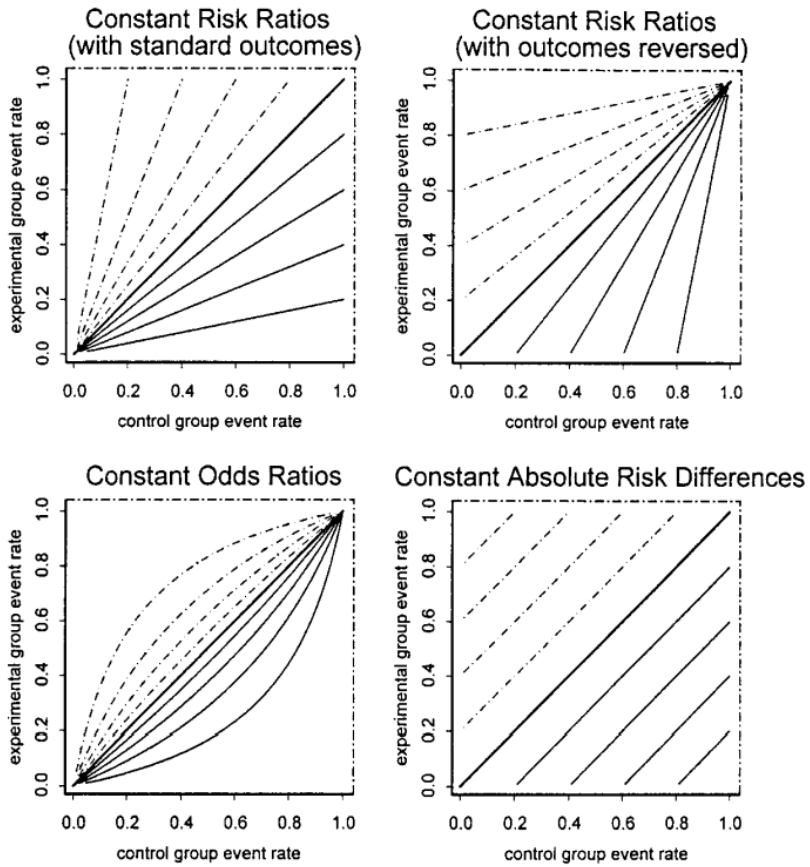
- Typically model summary statistics,
  - If individual-level data available can also do a “2-stage” approach; can help in obtaining more precise estimators from different studies
  - Fully joint model also possible: propagation of uncertainty

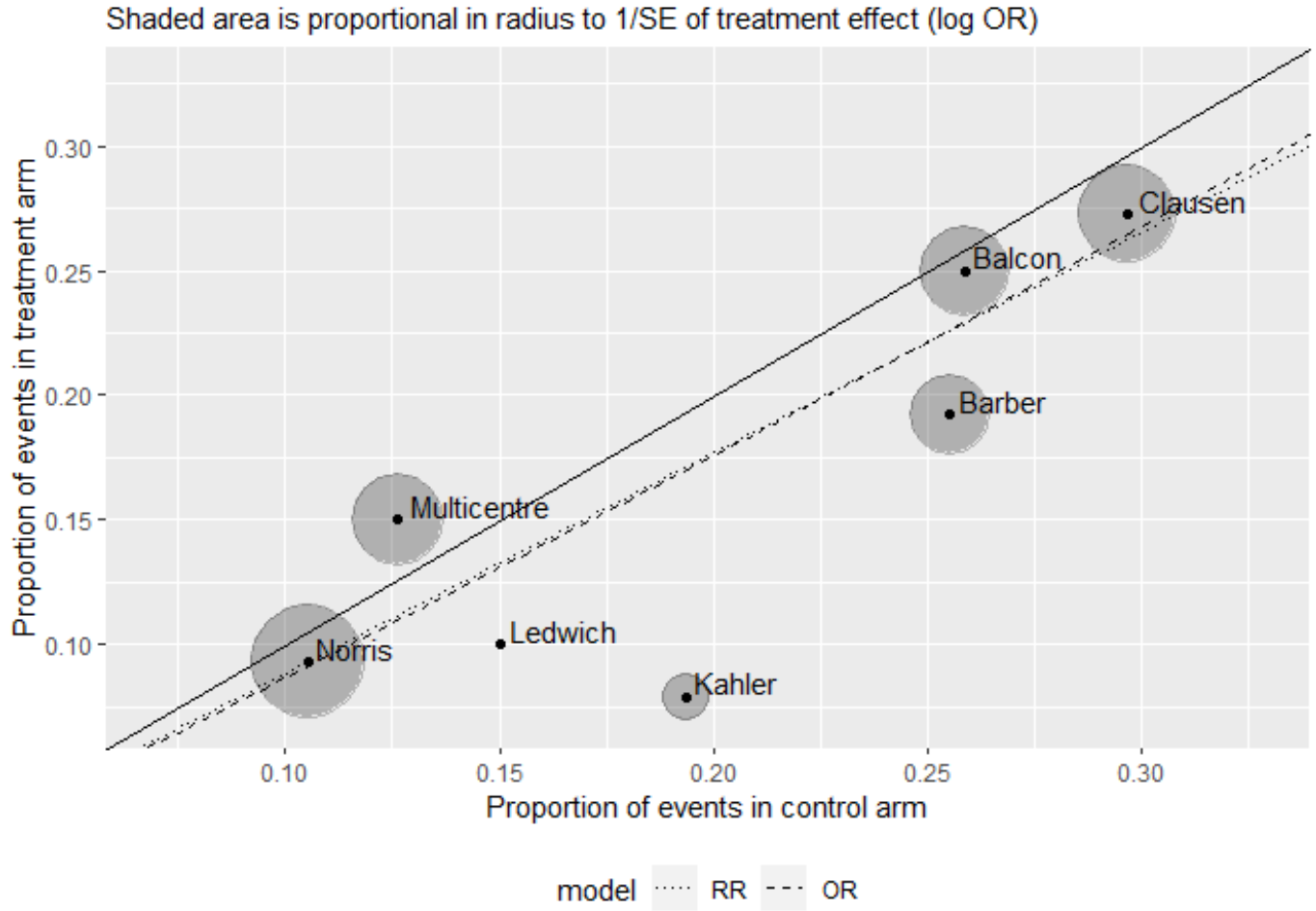
Role of graphical exploration

Not limited to forest plots

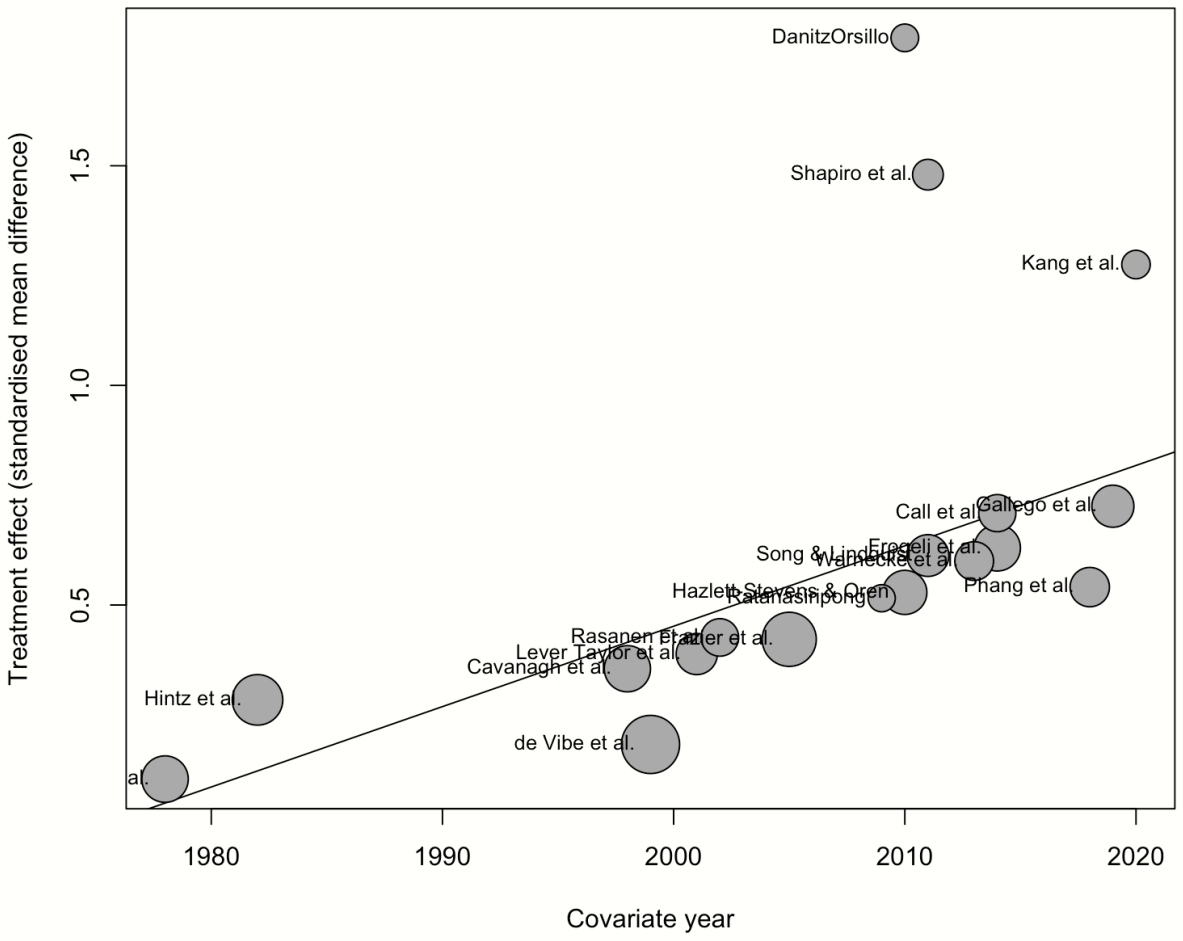
- Bubble plots: should covariates be included?
- L'Abbe plots: what (binary) outcome to use?
- Model comparisons and robustness checks: plot differences between individual studies, mean effects, predicted effects
- Plotting model vs data

Graphical exploration: examples





Graphical exploration: examples



Role of simulation

- Robustness checks: resimulate studies and fit candidate models
- Publication bias: add studies to the dataset
- Power calculations
- (For Bayesian models only) posterior predictive checking

Choice of model

- (Typically) fixed vs random effects (partial vs full pooling) + inclusion of covariates
  - FE vs RE terminology can be misleading and sometimes conflicting [1]
  - Good to consider: (1) Data generating process, (2) Predictive checks (can the model replicate data?), (3) Graphical comparison, (4) Formal model comparison (LOOCV)
  - …arguably easier in Bayesian models
- Bayesian vs frequentist estimate?

[1] Andrew Gelman "Why I don't use the term "fixed and random effects" | Statistical Modeling, Causal Inference, and Social Science (columbia.edu)

Method of inference

- Bayesian methods more technically demanding than frequentist
- Some cases where Bayesian models are especially useful:
  - Regulating amount of pooling with priors, informative priors
  - Imposing distribution of random effects (e.g. skewed)
  - (Subjectively) easier to compare many models
  - Using models as inputs into decision making (propagating uncertainty)
  - Jointly modelling individual-level data
  - Other models with complicated data generating processes

Making meta-analyses reproducible

- In R:
  - Frequentist: *metafor*, *meta*, *netmeta*
  - Bayesian: *baggr*
- Recent software advances make Bayesian inference for meta-analysis models as user-friendly as frequentist models
- Meta-analyses should
  1. publish code (data management, model fitting, plots)
  2. include raw data
  3. show data in the publication (table and/or plots)

Other forms of meta-analysis

Non-exhaustive list of topics we haven't covered:

- Meta-regression / mixed models
- Combining individual-level and summary data
- Network meta-analysis (indirect evidence)
- Impact on multiple parameters: mean & variance, quantiles, mixtures
- Dealing with small samples
- Model-based meta-analysis
  - Parametric survival curves

Good to be familiar with at least a couple, even as illustrations of the topics we discussed.

Summary

- Simulate data
- Explore graphically
- Compare many models